

# **Estimation of Housing Rental Prices in the USA Based on Structural and Descriptive Details**

ST 4035 – Group Project Proposal – Group E

| Section                          | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <b>Introduction</b>              | <p>Housing affordability is a critical concern in many parts of the United States, with rent prices varying widely across cities and states. To better understand the factors that influence rental prices, this project utilizes a dataset containing information on 10,000 apartment listings across the USA.</p> <p>The dataset includes 22 variables describing each listing, such as structural and descriptive details (bathrooms, bedrooms, square footage, amenities), location information (city, state, latitude, longitude, address), and listing metadata (category, title, source, whether photos are included, and pet policies). The target variable is price, representing the actual rental price of the apartment.</p> <p>The dataset originates from web-scraped rental listings and was later curated and hosted on the UCI Machine Learning Repository.</p> <p>Dataset: <a href="#">Apartment for Rent Classified - UCI Machine Learning Repository</a></p> |
| <b>Significance of the study</b> | <p>This study provides valuable insights into the property and listing factors that influence apartment rental prices, using data from the UCI “Apartment for Rent Classified” dataset. By applying statistical and machine learning techniques, it demonstrates how data-driven approaches can support accurate rental price prediction, helping tenants make informed decisions and landlords set fair market rates. The analysis also addresses key challenges such as handling categorical and text-based features, enabling better model interpretability and practical application in real estate platforms. This makes the study relevant for both real-world applications in housing markets and academic research in predictive analytics.</p>                                                                                                                                                                                                                          |
| <b>Objectives</b>                | <ul style="list-style-type: none"><li>● To explore the dataset to uncover key factors influencing rental prices.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

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|                    | <ul style="list-style-type: none"> <li>● To develop and evaluate predictive machine learning models for rental price estimation.</li> <li>● To integrate Explainable AI (XAI) techniques for model transparency and trustworthiness.</li> <li>● To design and implement a web application that delivers model predictions and explanations.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Methodology</b> | <ol style="list-style-type: none"> <li><b>1. Data Preprocessing</b> <ul style="list-style-type: none"> <li>● <b>Text Feature Extraction</b> (from title &amp; body): <ul style="list-style-type: none"> <li>○ Rent amount, bedrooms/bathrooms, unit type, location, amenities, lease terms</li> <li>○ Techniques: regex, NER, keyword rules</li> </ul> </li> <li>● <b>Missing Value Imputation:</b> <ul style="list-style-type: none"> <li>○ From text-derived features</li> <li>○ Contextual imputation using city/state-wise mean/mode</li> </ul> </li> </ul> </li> <li><b>2. Data Transformation</b> <ul style="list-style-type: none"> <li>● Standardize/normalize numerical features</li> <li>● Encode and regroup categorical variables (e.g., rare → “Other”)</li> </ul> </li> <li><b>3. Modeling</b> <ul style="list-style-type: none"> <li>● <b>Regression Models:</b> <ul style="list-style-type: none"> <li>○ Linear Regression, Regression Trees, Random Forest, XGBoost</li> </ul> </li> <li>● <b>Model Selection:</b> <ul style="list-style-type: none"> <li>○ Evaluate via cross-validation (e.g., 5-fold), RMSE as metric</li> <li>○ Hyperparameter tuning for Random Forest &amp; XGBoost</li> </ul> </li> </ul> </li> <li><b>4. Model Explainability</b> <ul style="list-style-type: none"> <li>● Use <b>SHAP</b> and <b>LIME</b> for feature impact analysis and local/global interpretability</li> <li>● Analyze feature importance for insight and transparency</li> </ul> </li> <li><b>5. Web App Development</b> <ul style="list-style-type: none"> <li>● Build an interactive app (Streamlit/Flask/Dash)</li> <li>● Predict rents, suggest alternatives, and visualize explanation.</li> </ul> </li> </ol> |